



B.TECH. COMPUTER SCIENCE AND ENGINEERING 2022

Module -2

[09 hours]

Properties of regular languages- proving languages not to be regular, the pumping lemma for regular languages , Closure properties of regular languages – closure of regular languages under Boolean operations-union, complementation, intersection & difference, Equivalence and minimization of automata

Module -3

[09 hours]

Context free grammars and languages: Definition of context-free grammars, derivations using a grammar, leftmost and rightmost derivations, the language of a grammar and sentential forms, exercise problems, Parse trees – constructing a parse tree, the yield of a parse tree, inference, derivations and parse trees, Applications of context free grammars.

Module -4

[09 hours]

Pushdown automata: The languages of a PDA – acceptance by final state, acceptance by empty stack Informal introduction, formal definition of PDA, a graphical notation for PDA's, The languages of a PDA – from empty stack to final state, from final state to empty stack Equivalence of PDA's and CFG's – from grammars to PDA, from PDA's to grammars

Module -5

[09 hours]

Introduction to Turing machines: Problems that computer cannot solve, The Turing machine – the quest to decide all mathematical questions, notation for the Turing machine, instantaneous descriptions for Turing machines. Programming techniques for Turing machines.

TEXT BOOKS

1. Introduction to Automata theory, Languages and Computation, John E. Hopcroft, Rajeev Motwani Jeffery D. Ullman, Third Edition Pearson education. 2007

REFERENCES

1. An Introduction to Formal languages and Automata, Peter Linz, III Edition